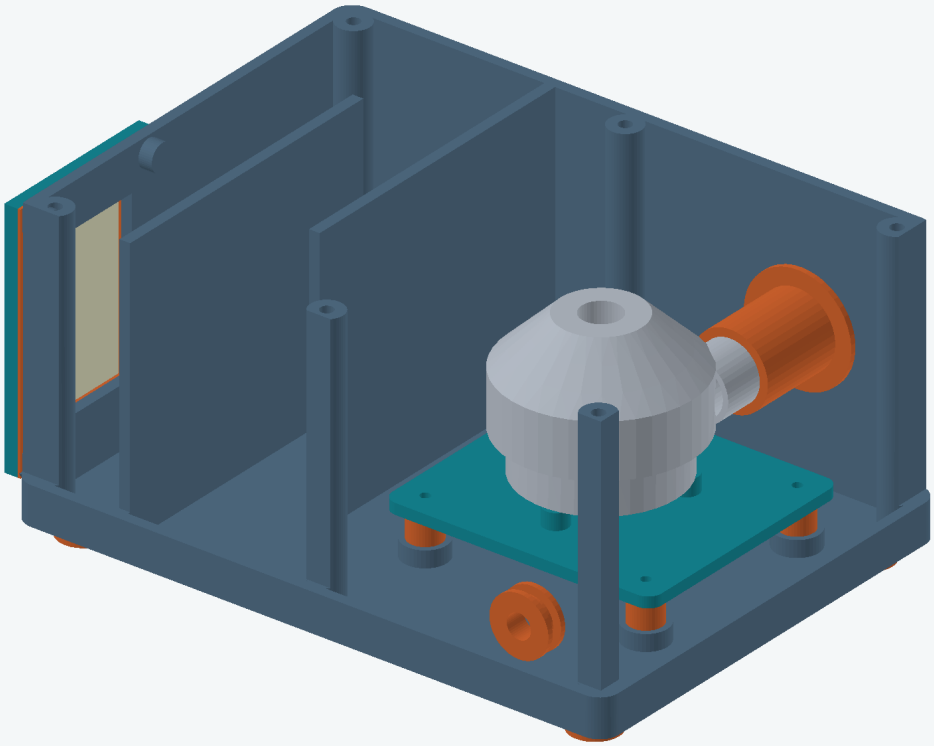
A quieter enclosure for WM6850

A serviceable 3D-printed concept for a CPAP blower. Acoustic, thermal and breathing-path performance remain unverified.

Inside the quiet enclosure

Lid removed; front and right walls cut away for visibility.



REV A / BENCH PROTOTYPE / SUPPLIER BLOWER SHAPE APPROXIMATE

220 x 160 mm body / 118 mm overall height / 4 printed parts

1 Redirect intake sound

Two alternating baffles remove a direct sight line to the blower. A gasket seals both baffle tops to the lid.

2 Isolate the blower

Four soft mounts support a separate carrier. A flexible hose and slack cable limit vibration transfer to the box.

3 Dampen the lid

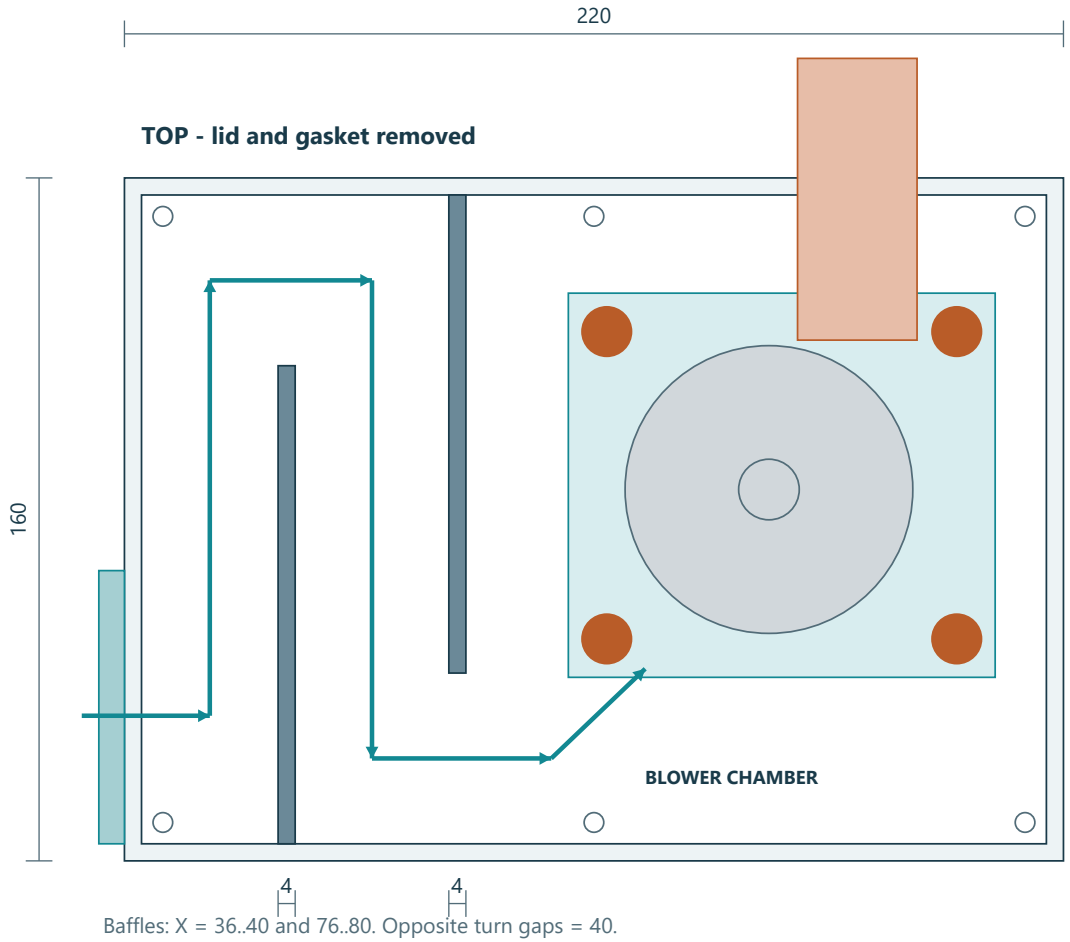
A metal cap over an external viscoelastic layer adds mass and damping. No exposed acoustic foam is specified.

CAD handoff: STEP assembly, individual STEP solids, printable STL files, FreeCAD project and regeneration source. Native SolidWorks files require conversion in SolidWorks; that application was not available for verification.

Before fabrication: verify the actual blower mounting holes, port fit, screw depth and wire exits. This design is a bench prototype, not a validated patient-use assembly.

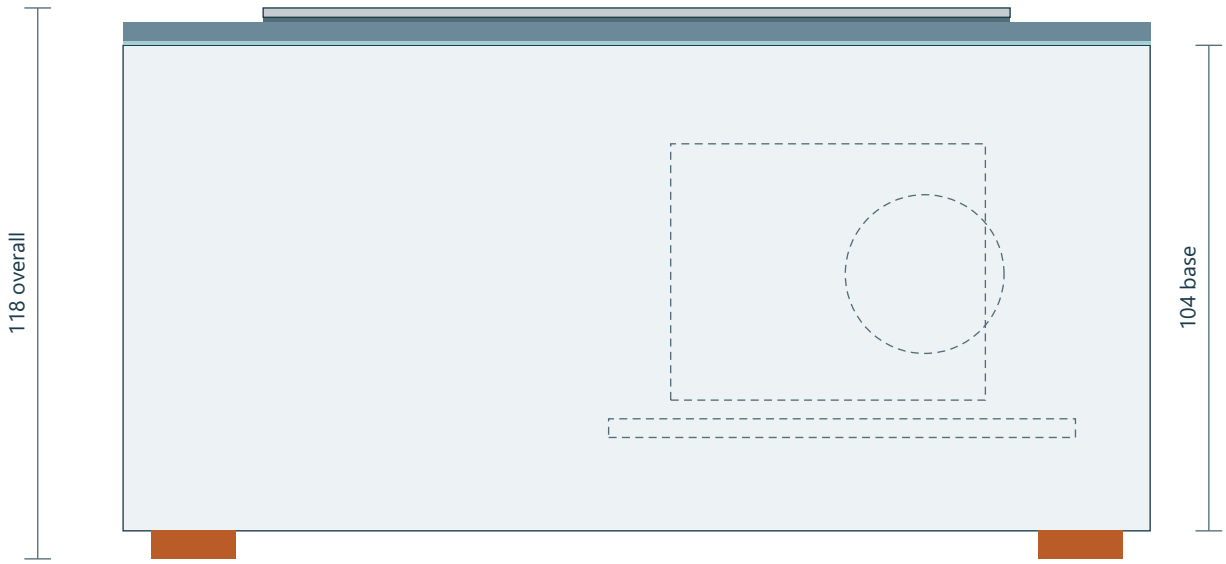
Overall arrangement

All dimensions in mm. Schematic orthographic arrangement derived from the CAD parameters; not a production tolerance drawing.



Baffles: X = 36..40 and 76..80. Opposite turn gaps = 40.

Arrows indicate intake route only; the blower shape is approximate.



FRONT - internal reference envelope shown dashed

Datum: X=0 left wall, Y=0 cable side, Z=0 underside of base. Lid underside Z=105; blower reference top Z=82.9. External hose/filter/gland envelope is 226 x 190 x 118 mm. Leave additional service space.

Interfaces to verify before printing

The supplier drawing fixes the main envelope. Items marked PROVISIONAL must be checked against a physical blower or supplier CAD.

Interface	Model value / location	Status
Blower envelope	Diameter 67.41; overall height 54.90; motor diameter 56	Datasheet p.5
Mounting	3 x M3, 120 degrees apart	Datasheet p.5
Carrier hole pattern	Diameter 50 PCD; angles 36, 156, 276 deg	PROVISIONAL
Blower mount plane	Centre X151, Y87; mount plane Z30; boss extends to Z28	Design placement
Outlet fitting	Barrel OD21.90; lip OD24; tip 44 from blower centre	Datasheet p.5
Outlet placement	Axis X171.7, Z55; tip Y131; bore diameter 18	Offset/bore PROVISIONAL
Wall passthrough	Diameter 34 at X171.7, Z55 in Y=160 wall	Design dimension
Carrier isolators	X/Y: (113,52), (195,52), (113,124), (195,124); diameter 12 x 10	Generic purchased envelope
Filter media	44 x 70 x 2; 36 x 62 window with two 3 mm bars	Media supplier TBD
Cable opening	Diameter 16 at X190, Z30 in Y=0 wall; split gland bore 9	Check actual harness
M3 insert pockets	Diameter 4.6 x 6 deep; 3.5 screw clearance	Match chosen insert

Carrier fit check

Print the carrier first. Match all three blower holes without force, check the central boss clearance and confirm screw engagement. The CAD does not establish the motor thread depth.

Outlet and intake fit check

Test a hose sample on the 24 mm lip. The top opening, volute contour and wire exits are simplified reference geometry; replace the blower envelope when a supplier STEP is available.

Do not infer a patient connector from the hose size. The outlet ends as a plain flexible tube. The existing CPAP interface, sensors and protective functions require their own specified fittings and system verification.

Parts and fabrication

Print one of each of the four STL files. Gaskets, vibration mounts, filters and hardware are separate purchased or fabricated items.

Item	Qty	Specification / action
Base with labyrinth	1	Print upright; 220 x 160 x 104. Local support at wall ports.
Lid	1	Print flat; 220 x 160 x 4. Six M3 clearance holes.
Blower carrier	1	Print flat; 100 x 90 x 4 plus three 6 mm spacers. Fit first.
Filter retainer	1	Print in supplied STL orientation; 64 x 90 face.
Rim/baffle + filter gaskets	1 each	Cut nonporous silicone. Rim: 1.5 free / 1 fitted; frame: 1 fitted.
M3 bonded soft mounts	4	Nominal diameter 12 x 10. Select stiffness for actual load.
Flexible outlet sleeve	1	Nominal 22 ID / 28 OD x 66; fit-check blower lip and clamps.
Outlet grommet / split gland	1 each	Custom soft silicone: 34 hole / 28 bore; 16 hole / 9 bore.
Intake filter	1	44 x 70 x 2 envelope; choose verified low-loss media and edge seal.
External damping laminate	1 set	160 x 110 x 1 viscoelastic sheet + 160 x 110 x 2 aluminium.
External feet	4	Diameter 18 x 6 nominal adhesive elastomer feet.
M3 heat-set inserts	14	6 lid + 4 mount bases + 4 filter. Match pilot and depth.
M3 screws, nuts, washers	See note	6 lid, 4 filter, 3 blower screws; 4 carrier nuts + washers.
Hose clamps / strain relief	As needed	Select after fit checks; envelopes omitted from CAD.

Bench-print starting point: PETG or ASA, 0.2 mm layers, 5-6 perimeters, 6 top/bottom layers, 40-60% infill. A brim needs a bed larger than the 220 mm part length. Printed volume is about 770 cm3 before slicer effects; the aluminium plate is about 95 g.

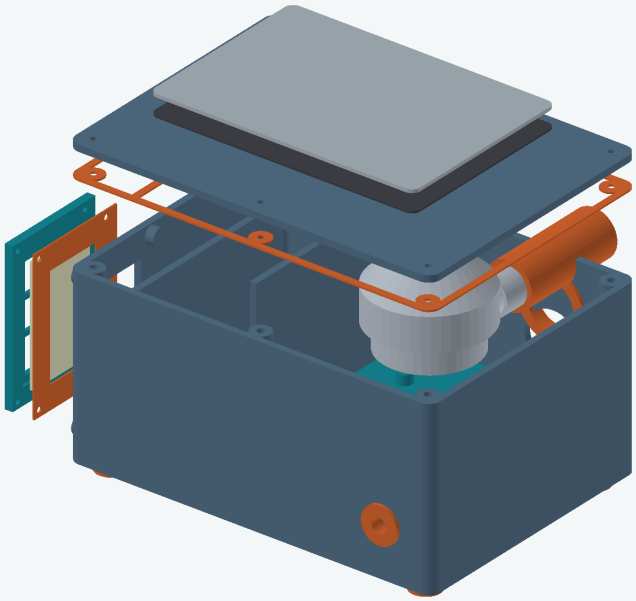
Hardware lengths: M3 x 10 is only a lid starting point. Choose every screw/stud length from actual washer stack, insert and motor thread depths. The carrier-to-blower screw path is 10 mm before thread engagement. No clamp, insert or thread is modelled.

Assembly sequence

Use the exploded view for component order. Locations are fixed in the neutral assembly; hardware is deliberately omitted.

Assembly arrangement

Components separated vertically for clarity; hardware omitted.



REV A / BENCH PROTOTYPE / SUPPLIER BLOWER SHAPE APPROXIMATE

01 Check the interfaces

Fit the carrier and hose sample to the actual blower. Update the model before printing the box.

02 Install the floating carrier

Fit base inserts, four isolators, carrier and blower. No rigid attachment should bypass the isolators.

03 Connect outlet and harness

Fit hose/grommet and split cable gland. Add retention, strain relief and an internal slack cable loop.

04 Seal the intake

Fit specified filter media, seal its edge and screw down the gasketed filter retainer.

05 Close and damp the enclosure

Fit the rim/baffle gasket and lid, tighten evenly, attach external damping laminate and feet.

06 Verify on the bench

Check clearance, leakage, restriction, noise, temperatures and transient pressure response before release.

Validation status and next measurements

The files are ready for CAD review and fit prototyping. No acoustic attenuation, CFD result or temperature qualification is claimed.

Passed: geometry and exchange

22 valid single solids; no overlaps above 0.01 mm3; STEP re-import contains 22 valid solids. Four printed STLs are closed meshes. SolidWorks import and the optional native-conversion macro remain untested.

Measure: noise at matched delivered performance

Compare bare versus enclosed at the same flow and delivered pressure, using the same hose and load. Record LAeq and spectra at 1 m at inlet, front and outlet sides. A proposed development target is at least 5 dB(A) insertion loss at an agreed operating point; this is not a prediction. The datasheet 45 dB(A) applies to its specified 1 kPa / 4 mm-orifice setup.

Measure: restriction and pressure response

Grille open area: 2,016 mm2; nominal minimum labyrinth channel: 3,200 mm2. At 300 L/min, grille velocity is 2.48 m/s. Assuming K=10 gives about 37 Pa of geometric loss, excluding filter and blower-entry losses; K is only a sensitivity assumption. Test actual losses with new and loaded filters. Proposed characterization: 4/10/20 cmH2O at 30/60/120 L/min, higher flows only where available.

Measure: temperatures, seals and materials

Log motor NTC, housing, outlet air and plastic-boss temperatures to steady state. An all-heat-to-air sensitivity gives about 17 degC rise at 120 L/min and 40.8 W; it is not a thermal prediction. Check leaks, hose retention, filter bypass, loose particles and cleanability. The datasheet prohibits complete blockage beyond 3 minutes; do not use prolonged occlusion tests.

Required before patient use

Qualify the complete breathing-gas pathway and printed surfaces, including particulate and volatile emissions, cleaning and system performance. No foam is exposed, but that alone does not qualify the materials. Confirm application-specific thermal and pressure limits before testing.

Sources and exchange guidance

User-supplied WM6850-24V specification, issued 2024-06-12: p.3 ratings and noise conditions; p.5 outline drawing. Original location is recorded in README.md.

SolidWorks: importing STEP, IGES and ACIS files. Import as an assembly, verify mm units and save referenced native parts.

NIOSH: engineering noise controls. Supports enclosure and isolation principles, not design-specific attenuation.

FDA: recognized ISO 18562-3:2024. Evaluation of volatile emissions in breathing-gas pathways.